

Electrical Fuse Schedule (Implementation - Lynx Topology)

As-of date: 2026-08-10

Purpose: define each required fuse by circuit, protected conductor/device, holder/housing method, physical placement, and linked wire-gauge assumptions for the approved Phase 1 Lynx architecture with a battery-backed 12V bus and dedicated 48V secondary alternator branch.

Related docs:

- Canonical electrical/system baseline: docs/core/SYSTEMS.md
- Implementation topology and conductor map: docs/implementation/ELECTRICAL_overview_diagram.md
- Decisions/open items tracker: docs/core/TRACKING.md
- BOM source of truth: bom/bom_estimated_items.csv

Design Basis

- Topology: Victron Lynx Distributor M10 (LYN060102010) with 4 active fused 48V branches; Slot 4 feeds Orion input.
- Battery bank assumption: 3x 48V 100Ah batteries in parallel (3 separate battery-positive conductors leaving batteries).
- 12V distribution assumption: 12V fuse block used as the shared junction device (main + stud = source combine, integrated negative bus/main - = return), fed by Orion-Tr Smart 48/12-30 charger and a 12V 100Ah LiFeP04 buffer battery branch.
- Parallel-bank safety rule: use one Class T fuse per battery-positive conductor leaving the battery.
- Active branch devices on Lynx:
 1. MultiPlus-II 48/3000
 2. SmartSolar 150/45
 3. Dedicated 48V secondary alternator branch (Mechman/WS500 path)
 4. Slot 4 / F-05 — Orion-Tr 48/12-30 input through one 40A MEGA body-marked at least 58VDC ; standalone F-06 is retired.
- Current shop identification note: do not infer the purpose of the physically X-marked/misrated Lynx fuse from the X alone. Identify its slot. Slot 2 requires F-03 60A/80V MEGA for the MPPT; Slot 4 requires F-05 40A MEGA marked >=58VDC for Orion. No 32V -rated fuse is acceptable on an energized 48V branch.
- Alternator path lock for this pass:
 1. F-04 is locked to 200A/80V MEGA in Lynx Slot 3 at the house-bank/Lynx end of the alternator positive run. Mechman's published 48V Elite curve reaches about 145.7A ; the 125% fuse basis is about 182A , so 200A is the next standard size and remains within the installed 2/0 AWG conductor envelope.
 2. Obsolete pre-Mechman charger/fuse paths are removed from active architecture and primary layout planning.
 3. Confirmed PH-VAN harness uses one short red/black pair as combined regulator power and voltage sense. Treat former separate F-12 regulator-power and F-13 positive-sense functions as one 15A fused red lead from the F-04 alternator/load-side stud. Use one compact Eaton/Bussmann HEB-AA in-line holder with a Littelfuse KLKD015.T 600VAC/DC fuse; no DC fuse panel or DIN enclosure. Put the Wakespeed shunt on the battery side of the hard-mounted SmartShunt; its purple/high and grey/low sense leads remain unfused.
 4. Ford Upfitter Switch #3 is locked as the manual WS500 enable source through F-15 .

Alternator Branch Fuse + Wire Finalization (2026-03-19)

Assumptions for this pass:

1. One-way alternator run length: 20 ft .
2. Dedicated positive and dedicated negative runs (loop length 40 ft).
3. Charging target voltage basis: 58.4V .
4. Voltage-drop planning target: about <=2% under expected charging current.

Resistance and drop screen ($V_{drop} = I * (2 * L * R_{per_ft})$):

Gauge	150A drop	% @ 58.4V	200A drop	% @ 58.4V	Decision note
4 AWG	1.49V	2.55%	1.99V	3.40%	Reject for this run length/current class
2 AWG	0.94V	1.61%	1.25V	2.14%	Acceptable for ~150A class
1/0 AWG	0.59V	1.01%	0.79V	1.35%	Strong margin
2/0 AWG	0.47V	0.80%	0.62V	1.07%	Best margin; inventory already on hand

Lock for this build pass:

- Use F-04 200A/80V in Lynx Slot 3. This preserves margin above the published 145.7A curve while protecting the existing 2/0 AWG run.
- Reuse existing uncut 2/0 inventory for alternator + and dedicated negative run.
- Inactive BOM row 173 preserves the removed contingency estimate; open a new active line only if field measurement proves an actual shortfall.

Required Fuse Map (Start-to-Finish, With Housing)

Fuse ID	Circuit (source -> load)	Protected wire/device	Fuse type and voltage class	Amperage	Holder or housing method	Physical location
F-01A	Battery A + -> bank positive combine/disconnect input	Battery A positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment within ~`7"`` Battery A positive post
F-01B	Battery B + -> bank positive combine/disconnect input	Battery B positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment within ~`7"`` Battery B positive post
F-01C	Battery C + -> bank positive combine/disconnect input	Battery C positive cable leaving battery	Class T (>=125VDC)	200A provisional	Blue Sea Class T fuse block (covered stud mount, 110A-200A family)	Battery compartment within ~`7"`` Battery C positive post
F-02	Lynx Slot 1 -> MultiPlus DC+	Main inverter positive feeder	MEGA, 58V or 80V	125A	Integrated Lynx Distributor fuse slot	Lynx Distributor, S 1
F-03	Lynx Slot 2 -> SmartSolar BAT+	MPPT battery-side positive feeder	MEGA, 80V Victron replacement stock	60A	Integrated Lynx Distributor fuse slot	Lynx Distributor, S 2
F-04	Lynx Slot 3 -> 48V alternator branch input (ALT+)	Alternator-to-house positive charge cable	MEGA, 80V	200A	Integrated Lynx Distributor fuse slot	Lynx Distributor, S 3
F-05	Lynx Slot 4 -> Orion 48v input +	Orion 48v input feeder	MEGA 40A, body-marked >=58VDC; Victron CIP138040020 40A/80V replacement fallback	40A deliberate feeder-protection value; above Victron's 20A table recommendation but safely below installed 6 AWG ampacity	Integrated Lynx Distributor fuse slot	Lynx Distributor, S 4
F-06	Retired standalone Orion input fuse position	Not active	N/A	Not installed	No holder	N/A

F-07	Orion 12V output + - > 12V fuse block main + stud	Main 12V feeder from Orion into shared source-combine point	MEGA, 80V Victron replacement stock	60A	Victron MEGA fuse holder (external, non-Lynx)	Electrical cabinet, within ~`7" of Orion12V` output stud
F-09 (not presently required)	Selected Arch Pro 352P string positives -> branch pair or two-string combiner -> MPPT PV input	Add only if the received-module/code/backfeed review requires individual string OCP	Listed gPV, voltage class above the final 142.29V cold string calculation	No fuse in the current two-string design; if required later, size from received I _{sc} , 15A module series-fuse rating, conductor ampacity, and applicable PV rules	Compatible branch connectors or a listed two-string combiner; do not procure the retired three-position fuse package	Roof-entry/service area
F-10	12V fuse block branch circuits -> each 12V load	Individual 12V branch conductors and load circuits	ATO/ATC blade fuses (32V class)	Per-circuit	Integrated sockets in marine 12V fuse block	12V fuse block in electrical cabinet
AUDIO-HU / 12V-12	12V fuse block -> Kicker 46KMC2 media center/source unit	KMC2 source/head-unit branch conductor and device harness	ATO/ATC branch fuse (32V class) plus KMC2 harness 15A ATM fuse	15A	Integrated 12V fuse block branch plus OEM KMC2 harness fuse	Electrical cabinet to driver-side DC shelf/source face
AUDIO-SUB	12V source/main + stud -> external fuse -> Kicker 49PTRTP10 powered sub +	Powered-sub positive branch and subwoofer input; manual-required external protection	Inline/MRBF/AFS/ANL-class fuse matched to selected 4 AWG kit (32V+ DC class acceptable on 12V branch)	40A	Holder within about 18 in of 12V source takeoff; if using Kicker 47KMPK4, fit the PTRTP10-required 40A fuse rather than the kit generic larger fuse	Near 12V source/junction downstream of SW-12V-BATT
F-11	12V buffer battery + -> 12V fuse block main + stud via SW-12V-BATT	Buffer battery source cable and downstream junction fault exposure	Inline MIDI/AMI/ANL family rated >=32VDC	100A class baseline	Sealed inline holder mounted close to battery positive	Within ~`7" of 12V buffer battery positive post
F-12/F-13-PHVAN	F-04 alternator/load-side stud -> WS500 PH-VAN red lead	Combined WS500 regulator power and positive voltage-sense lead	Littelfuse KLKD015.T fast-acting midget fuse, 600VAC/DC	15A	Eaton/Bussmann HEB-AA water-resistant in-line holder, 600V	In-line immediately beside F-04; short 14 AWG pigtails with sealed step-down splice to the 16 AWG harness; no panel, DIN rail or enclosure
CERBO-PWR	48V system positive -> Cerbo GX power +	Cerbo GX low-current electronics feed	Inline fuse/holder rated for 48V bank maximum voltage	1A-3A	Small inline holder close to	Electrical cabinet, preferably

					the positive takeoff	system/load side of main disconnect so Cerbo powers down with the house system during bench shutdown
F-15	Ford Upfitter #3 -> WS500 brown ignition/enable wire	Low-current regulator enable/control wire	Inline ATC/ATO (32v class acceptable; 12V control circuit)	3A	Sealed inline holder near the Ford upfitter blunt-cut source / splice handoff	Engine bay or control-wire handoff point before small-gauge run to WS500
OEM-SHUNT	Battery-side positive or Lynx/system positive tap -> SmartShunt Vbatt+ terminal	SmartShunt electronics supply/sense lead	External Victron-supplied red cable with inline low-current fuse; not an internal SmartShunt fuse	OEM value	Inline holder in supplied red cable	Prefer battery side positive SOC continuity is desired when the main disconnect is open; system side is acceptable if zero parasitic draw during disconnect-of-storage matters more

Retired Fuse IDs

Obsolete pre-Mechman charger/fuse paths are removed from the active schedule. Do not reserve board space or labels for them.

Spare Fuse Inventory (Updated)

Fuse type	Installed qty	Spare qty to carry	Notes
Class T 200A (provisional installed)	3	1 on hand	Owner confirmed 3 holders and 4 slow-blow Class T fuses total (3 installed + 1 spare); add more only if the one-spare-per-installed policy is later desired
MEGA 200A/80V	1	1 minimum	Alternator branch (F-04); Amazon Victron stock is commonly sold as a five-pack, so install one and carry one while retaining the remainder as shop stock
MEGA 125A (58V/80V)	1	4	MultiPlus branch
MEGA 60A (80V)	2	3	Victron 5-pack row 188: install MPPT (F-03) + Orion output (F-07), keep 3 spares. Earlier 6x low-cost 60A MEGA batch was owner-confirmed misadvertised/not actually 58V and is quarantined/trash, not 48V install stock.
MEGA 40A, >=58VDC	1 active	1-2	Orion input in Lynx Slot 4 (F-05); existing body-marked 58VDC stock is acceptable under the locked 56.8v charge ceiling; use Victron CIP138040020 40A/80V as replacement fallback
Retired Orion standalone input-fuse stock	0 active	Optional purchased stock only	30A/58V MIDI and FKS/ATO parts are not installed; no USM1/ATM20 purchase required

12V buffer battery main fuse (100A class)	1	3	Spare pack basis is BOM row 105
WS500 PH-VAN combined regulator power / positive-sense fuse (F-12/F-13-PHVAN)	1 required position	1 optional later	One Littelfuse KLKD015.T 15A/600VAC/DC midget fuse in an Eaton/Bussmann HEB-AA in-line holder is sufficient to commission; no panel or proprietary automotive contacts are needed
Cerbo GX power fuse (CERBO-PWR)	1 active position	1	Carry a spare 1A-3A fuse/holder rated for the 48V bank maximum
WS500 ignition/enable fuse (F-15)	1 active position	2	Carry spare 3A mini/ATO fuse and one spare sealed holder; 12V control circuit
PV string OCP (F-09)	0 planned	0	Selected array has only two parallel 3S strings. One peer string cannot backfeed enough current to approach the Arch Pro 15A series-fuse rating. Reopen only if received labels, conductor/connector limits, or applicable PV rules require it.
SmartShunt OEM harness fuse	1	1	Keep OEM-equivalent spare if field-replaceable
ATO/ATC branch fuses	variable	2 each used value	Keep mixed kit onboard
Camper audio KMC2 branch (AUDIO-HU)	1 active 15A branch plus KMC2 harness 15A ATM	2 spare 15A blade/ATM fuses	Preserve both source-side conductor protection and the KMC2 harness fuse
Camper audio powered sub (AUDIO-SUB)	1 active 40A external fuse	1-2 spare 40A fuses matching the selected holder family	PTRTP10 manual calls for 40A external fuse and 4 AWG power/ground

BOM Row Mapping

Fuse scope	BOM row(s)
Main battery Class T protection (F-01A/F-01B/F-01C) + Class T spares	bom/bom_estimated_items.csv row 7
Lynx branch MEGA fuses (F-02 to F-05 installed) + spare set	bom/bom_estimated_items.csv rows 10, 170, 188, and 323
Orion installed fuse-holder hardware (F-05 Lynx input, F-07 external output)	bom/bom_estimated_items.csv rows 6 and 11
Retired Orion standalone input-fuse stock (F-06)	Purchased stock: active BOM rows 133, 182, and 321; inactive rows 326 and 230 preserve retired holder/bridge history. Row 321 empty FKS housings and row 133 20A fuses are not used anywhere in the final alternator/Orion topology.
WS500 regulator-power and voltage-sense protection (F-12/F-13-PHVAN)	Active BOM row 171; inactive row 320 preserves the retired separate-3A concept
Cerbo GX power feed (CERBO-PWR)	bom/bom_estimated_items.csv row 22; small inline fuse/holder may come from low-current install stock
WS500 Upfitter #3 enable/control path (F-15)	bom/bom_estimated_items.csv row 176
12V buffer battery (B12)	bom/bom_estimated_items.csv row 21
12V buffer battery main fuse + holder (F-11)	bom/bom_estimated_items.csv row 125

12V battery disconnect (SW-12V-BATT)	bom/bom_estimated_items.csv row 124
12V branch panel and blade fuses (F-10)	bom/bom_estimated_items.csv row 16
Camper audio source/head-unit branch (AUDIO-HU / 12V-12)	bom/bom_estimated_items.csv rows 189, 192, and 193
Camper audio powered-sub branch (AUDIO-SUB)	bom/bom_estimated_items.csv rows 191 and 192
SmartShunt external OEM fused red lead (OEM-SHUNT)	bom/bom_estimated_items.csv row 23 (included with SmartShunt kit)
Final PV string OCP / combiner / disconnect allowance (F-09, if required)	bom/bom_estimated_items.csv row 106

Assumptions and Open Items

- Wire sizing above assumes copper conductors and enclosed vehicle routing.
- Solar topology is locked to 6x BougeRV Arch Pro 100W , 3S2P , on the Victron 150/45 . Each string is 113.4V Voc and 3.2A Isc at STC; the tolerance-aware cold calculation is 142.29V at -40F . With only two parallel strings, no individual F-09 is presently planned. Verify the received module labels, connector/conductor ratings, moving-route ampacity, and final code basis before as-built release; do not purchase or install the retired 3x 15A gPV package.
- Do not complete the owned Littelfuse 178.6150.0001 empty housings. Use the Eaton/Bussmann HEB-AA + Littelfuse KLKD015.T in-line branch for the PH-VAN red lead. The Wakespeed shunt is on the battery side of the hard-mounted SmartShunt, uses no 5A sense fuses, and must be configured Shunt at Battery .
- SW-12V-BATT remains manual-only in Phase 1; no automatic LVD behavior is assumed.
- Final lock for F-11 still requires explicit 12V buffer battery/BMS continuous discharge-current confirmation.
- F-15 exists to protect the smaller-gauge control wire between Ford Upfitter #3 and the WS500 brown ignition/enable wire; the factory 25A upfitter circuit protection is not the final wire-protection device for that branch.
- Orion F-05/F-06 lock: the active topology uses F-05 , one 40A MEGA body-marked at least 58VDC in Lynx Slot 4, feeding the existing 6 AWG Orion input directly. F-06 is retired. This deliberately prioritizes simple feeder protection over strict adherence to Victron's 20A table recommendation; do not stack an inline fuse after the Lynx fuse.
- Camper audio 12V load: KMC2 source branch and PTRTP10 powered-sub branch are downstream 12V loads, not new 48V branches. The PTRTP10 40A branch can exceed Orion 30A continuous output during heavy bass; rely on the 12V buffer battery for transients and validate sustained loud-use behavior before assuming all 12V loads can run at max simultaneously.
- Orion holder correction: no standalone input holder is required. The purchased 178.6150.0001 housing-only parts and 30A/58V MIDI stock remain unused; do not salvage or stack them into the final branch. Torque the Slot 4 MEGA hardware and Orion terminals to current manufacturer values, tug-test, and strain-relieve the existing 6 AWG pair.